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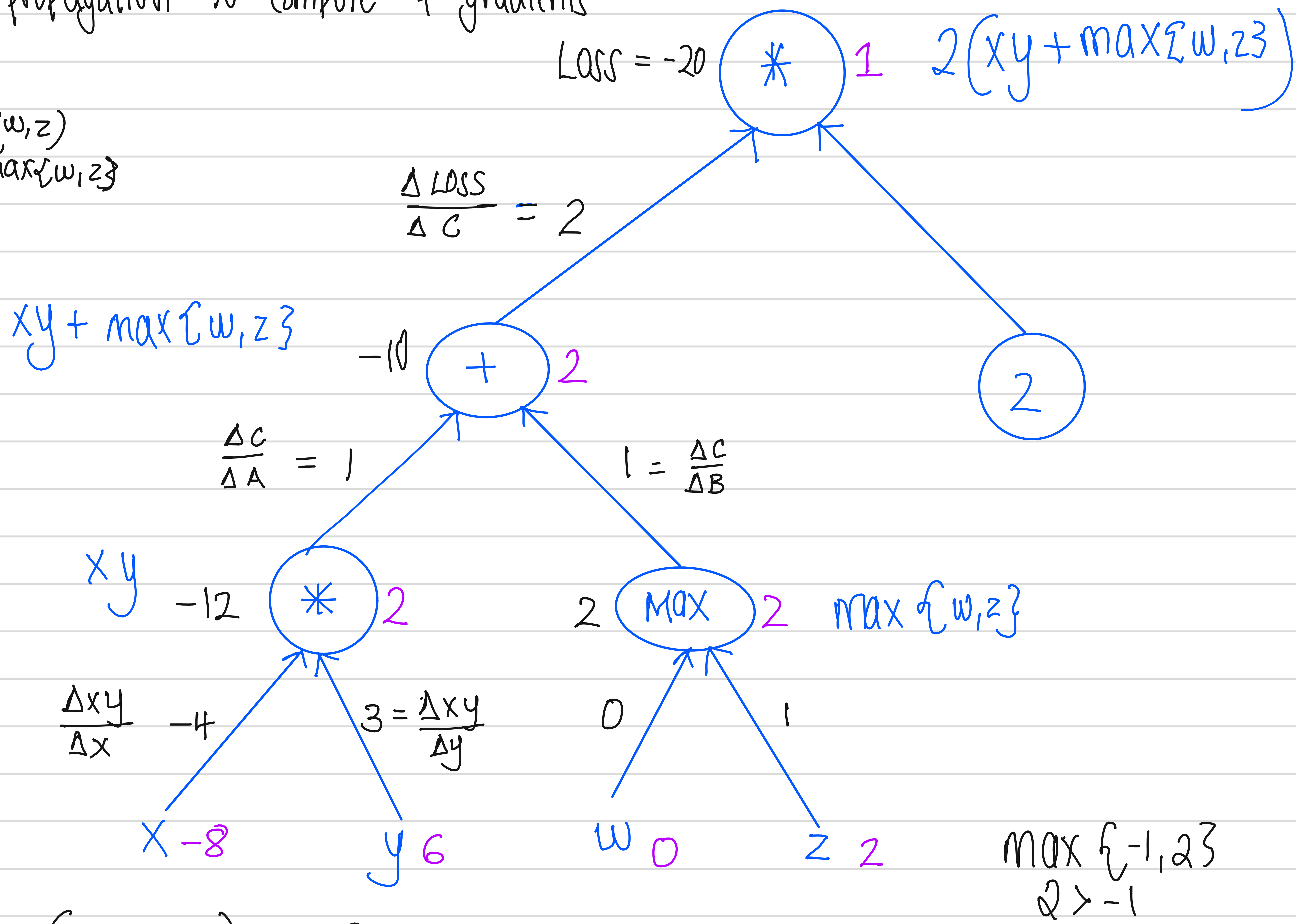
Assignment 4

1) $Loss(x, y, z, w) = 2(xy + \max\{w, z\})$

Back propagation to compute 4 gradients

$A = xy$
$B = \max\{w, z\}$
$C = xy + \max\{w, z\}$



$x = 3$
$y = -4$
$z = 2$
$w = -1$

$\nabla_x Loss(x, y, w, z) = 2 \cdot 1 \cdot -4 = -8$

$\nabla_y Loss(x, y, w, z) = 2 \cdot 1 \cdot 3 = 6$

$\nabla_w Loss(x, y, w, z) = 2 \cdot 1 \cdot 0 = 0$

$\nabla_z Loss(x, y, w, z) = 2 \cdot 1 \cdot 1 = 2$

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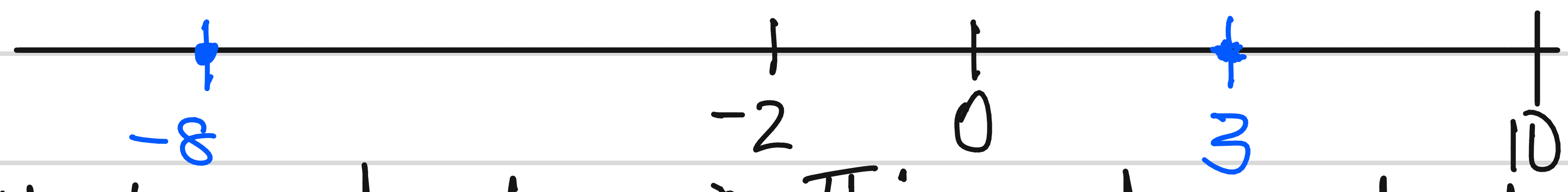
2) K-means clustering with $K=2$ on $\{-2, 0, 10\}$

$u_1 \in R$

$u_2 \in R$

$u_1 < u_2$

Given assumption 2: If no points are assigned to a cluster j, the centroid j is set to ∞. Suppose 2 centroids were selected $u_1 = -8$ and $u_2 = 3$. In this case, all the 3 points will be assigned to cluster 2 using u_2 centroid, as $-2, 0, 10$ are close to 3 compared to 8.



In the next step, centroid u_1 will be set to ∞. This gets stuck in a local optima.

Using another example, $u_1 = -3$, $u_2 = 5$, -2 and 0 will be assigned to cluster one with centroid -3. While 10 is assigned to cluster 2 with $u_2 = 5$. In the next step:

Using points, I change centroids. For cluster 1, $(-2+0)/2 = -1$, $u_1 = -1$ and cluster 2 since 10 is the only point, $u_2 = 10$.

From this I understand that in order to avoid having a cluster with no points, u_1 has to be closer to the lowest value, -2, and u_2 has to be closer to biggest value, 10.

$$(u_1 - 2)^2 < (u_2 - 2)^2 \quad \text{And} \quad (u_1 - 10)^2 > (u_2 - 10)^2$$

This ensures, I have atleast 1 point in each cluster at the start. This avoids assigning ∞ to centroids.

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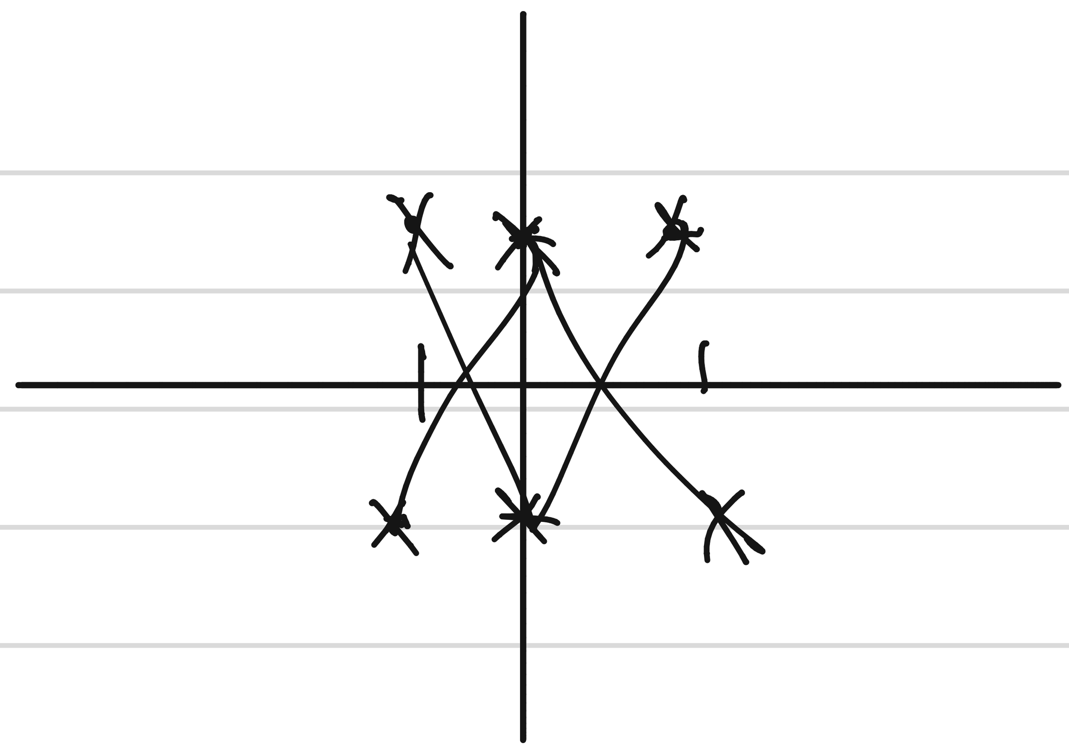

3). 2 training datasets

$D_1 = (-1,+1),\ (0,-1),\ (1,+1)$

$D_2 = (-1,-1),\ (0,+1),\ (1,-1)$

Define 2d feature function $\phi(x)$

- weight vector ω_1 s.t: $\omega_1 \cdot \phi(x) > 0$ if $x = +1$
  $\omega_1 \cdot \phi(x) < 0$ if $x = -1$

- ω_2 classifies D_2 perfectly.

So, it is not possible to separate $+1$s and -1s in 1D. For example in D_1 there is a point $(0,-1)$, negative y, between points $(-1,+1)$ and $(1,+1)$

$\phi(x) = x$ fails

2d Try $\phi(x) = [1, x^2]$ Decided to go with x^2 to remove the negatives, also this ensures feature vectors are the same.

Datasets:

$$D_1 = \{([1,1],+1),\ ([1,0],-1),\ ([1,1],+1)\}$$

$$D_2 = \{([1,1],-1),\ ([1,0],+1),\ ([1,1],-1)\}$$

For ω_1:

$w_1 \cdot \phi(x) > 0$ when $y = +1$

$w_1 \cdot \phi(x) < 0$ when $y = -1$

In D_1

$(-1,1)$

$w_1 \cdot [1,1] > 0$

$w_a \cdot 1 + w_b \cdot 1 > 0$

$w_a + w_b > 0$

$(0,-1)$

$w_1 \cdot [1,0] < 0$

$w_a \cdot 1 + w_b \cdot 0 < 0$

$w_a < 0$

$(1,+1)$

$w_1 \cdot [1,1] > 0$

$w_a \cdot 1 + w_b \cdot 1 > 0$

$w_a + w_b > 0$

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$w_a < 0$ and $w_a + w_b > 0$

$w_a = -1$ then $-1 + w_b > 0$ then $w_b = 3$

$$w_1 = [-1, 3]$$

Using w_1 with any $\phi(x)$ from D_1 will give the correct y. For example point $(-1, +1) \Rightarrow \phi(x) = [1, x^2] = [1, 1]$

$$w_1 = [-1, 3]$$

$$[-1, 3] \cdot [1, 1] = -1 \cdot 1 + 3 \cdot 1 = -1 + 3 = 2 > 0.$$

and $y = +1$

For w_2:

$w_2 \cdot \phi(x) > 0$ when $y = +1$

$w_2 \cdot \phi(x) < 0$ when $y = -1$

In D_2 we have:

$(-1, -1)$	$(0, +1)$	$(1, -1)$
$w_2 \cdot [1, 1] < 0$	$w_2 \cdot [1, 0] > 0$	$w_2 \cdot [1, 1] < 0$
$w_a + w_b < 0$	$w_a > 0$	$w_a + w_b < 0$

$w_a > 0$ and $w_a + w_b < 0$

Say $w_a = 2$ then $2 + w_b < 0$ then $w_b = -3$.

$w_2 = [2, -3]$

Using w_2 with any $\phi(x)$ from D_2 will give the correct y. For example point $(-1, -1) \Rightarrow \phi(x) = [1, 1]$

$w_2 = [2, -3]$

$$[2, -3] \cdot [1, 1] = 2 \cdot 1 + -3 \cdot 1 = 2 - 3 = -1 \quad \text{and } y = -1$$

So $w_1 = [-1, 3]$ and $w_2 = [2, -3]$

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General Question 3). Is every dataset linearly separable in some feature space?

Yes, you can create a feature vector $\phi(x)$, such that for every x in dataset it produces correct y. BUT, the example given above, Datasets 1 and 2 only had 3 x,y pairs. And I had to go through each in order to determine weight vectors w_1 and w_2.

So, its possible but you have to go through the whole dataset. Additionally, it may only work for the pairs found in the given dataset. So it may work for training, but may not work for a pair that does not exist in the dataset.

According to me, this is not going to be a good feature extractor to use, because it may not work correctly using pairs never seen before.